StoreVisionPro

Unified Requirements and Product Blueprint

AI-powered shop analytics and marketing for any retail business

About this document

This version merges the two source files into one document: the product vision, flow, schema, UI blueprint and roadmap from **StoreVisionPro2.pdf**, and the formal requirements from **requirements_updated_merged_with_additions.md** (Requirements 1 to 24): AI governance, authentication, multilingual, offline-first, market intelligence, expiry management, branding onboarding and QR-menu ordering. Every requirement carries a **Source** tag so you can see where it came from.

Both	In both files	**MD**	Only in the requirements MD	**PDF**	Only in StoreVisionPro2.pdf

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1. Introduction and Vision

StoreVisionPro is a mobile (Android first) and later desktop application that any shop owner (general store, grocery, clothing, electronics, cosmetics and more) can use to:

- Track products, customers, sales, expenses and suppliers
- Analyse performance, including peak times, best and worst sellers, and profit
- Predict trends, demand, stock-outs and seasonal peaks
- Generate marketing content automatically (posters, captions, festival wishes, ads)
- Improve decisions with an AI Business Health Check and a chatbot / voice assistant
- Keep working offline, in the shop owner's language, with the owner always in control of AI suggestions

Six product modules

Module	Purpose
Module 1	Shop Operations (Sales, Products, Inventory)
Module 2	Analytics Dashboard
Module 3	AI Business Insights
Module 4	AI Chatbot and Voice Assistant
Module 5	AI Marketing Creator
Module 6	Business Management Tools

Guiding principles

- **Owner in control.** AI only suggests. The owner accepts, edits or rejects; nothing is applied or published automatically.
- **Offline-first.** Data is stored locally (Room / SQLite) and is never lost during network or server failure.
- **Honest AI.** If data is missing or insufficient, the system says so instead of inventing an answer.
- **Evidence-based AI.** Market trends and price comparisons show their source and date; nothing is invented or presented as fact without support.
- **Industry-aware.** Recommendations adapt to the shop type chosen at registration.
- **Simple for small shops.** Fast entry, barcode scanning, voice input, local language and clean dashboards.

2. Glossary

Term	Definition
Shop Owner	Primary administrator of the business. Final authority over all AI suggestions.
Staff User	Employee with limited access: entry-only, view-only or edit-level.
POS System	Point-of-sale functions: recording sales and managing products and inventory.
Analytics Engine	Processes sales data into insights, reports and visualisations.
Inventory Tracker	Monitors stock levels and generates low-stock and reorder alerts.
AI Marketing Generator	Generative AI that creates posters, captions and festival creatives for Instagram, WhatsApp and other platforms, using business performance data or the owner's chosen offer. Festival posters carry the business logo/name and social media IDs.
Business Health Check	AI assessment of business health that suggests strategies, affordable offers and schemes, considering budget, profit/loss margin, customer benefit and the target customer group.
AI Chatbot	Conversational assistant that answers questions about stock, sales and profit, runs the Business Health Check on request, and proposes updates for confirmation.

Term	Definition
Voice Log	Agentic AI feature that converts spoken input into a structured proposed record for review.
PDF Export	One-click PDF of daily, weekly, monthly or yearly reports for printing.
Offline Mode	Local storage (Room / SQLite) that lets the app work without internet and sync later.
Firestore Database	Cloud NoSQL database for application data.
Market Intelligence	Externally sourced trend and price information shown with source and date, always kept separate from the shop's own records.
Near-Expiry Stock	Stock whose recorded expiry or best-before date falls inside the owner's configured warning window.
Branding Profile	Per-shop store of the approved logo, logo versions, colours and branding preferences reused by marketing tools.
QR Menu	QR code that opens a hotel, restaurant or cafe menu where customers can place orders.

3. Functional Requirements by Module

Keywords SHALL (mandatory), SHOULD (recommended) and MAY (optional) indicate priority. Source tags: Both, MD, PDF.

Module 1: Shop Operations (Sales, Products, Inventory)

User story: As a shop owner, I want to record products, sales, stock, expenses and suppliers with accurate date and time, so that I have reliable business records to analyse.

ID	Requirement	Source
1.1	Product management. The system SHALL let an authorised user add a product with name, category, price and initial stock quantity (all required), plus an optional photo, and SHALL allow editing and deleting products.	**Both**
1.2	Barcode / QR scanning. WHEN a user scans a barcode, the system SHALL capture the product details (name, image, price where available), pre-fill the product form and allow fast product lookup during sales entry.	**Both**
1.3	AI-assisted product entry. WHEN the owner captures or uploads a product image, an AI product-recognition component SHALL suggest name, description, category and price using image and (where available) internet information. Suggested values SHALL be shown as editable suggestions; only values explicitly confirmed by the user SHALL be saved.	**MD**
1.4	Initial stock rule. An initial stock quantity SHALL be derived only from owner-supplied information or existing records. An internet lookup SHALL NOT be treated as authoritative stock data.	**MD**
1.5	Sales entry. The sales screen SHALL provide product search/dropdown, quantity stepper, display of current stock, a running bill summary with auto-calculated total, customer count, and a date/time field that is auto-filled with the current time and editable.	**Both**
1.6	Sale items. WHEN multiple products are sold together, the system SHALL create individual sale items linked to one main sale record, and SHALL persist the sale to the database immediately after completion (to the local store first when offline).	**Both**
1.7	Calendar view. The system SHALL provide a calendar to navigate sales day-wise and month-wise, and selecting a date SHALL list all sales made on that day.	**PDF**
1.8	Customer visit tracking. Each sale SHALL store a customer count (default 1). The system SHOULD later support a separate 'visited, no purchase' counter.	**PDF**
1.9	Real-time inventory. WHEN a product is sold, stock SHALL reduce automatically by the sold quantity. The system SHALL raise low-stock alerts (custom threshold per product), stock-out notifications, and pre-peak reminders such as 'add stock before the weekend peak'.	**Both**
1.10	Expense tracking. The system SHALL record expenses categorised as rent, electricity, salary or stock purchase, show a monthly expense summary, and accept entry by manual input or Voice Log.	**Both**
1.11	Supplier management. The system SHALL store suppliers with contact details, the supplier for each product, purchase cost, delivery schedule and next expected delivery date, supplier-wise history, and supplier payment tracking with due dates and payment history.	**Both**
1.12	Users and roles. The Shop Owner SHALL have full access. Staff accounts SHALL be assigned one of: entry-only, view-only, or edit-level manager. Role changes SHALL take effect immediately across all active sessions, restricted actions SHALL be blocked, and all user actions SHALL be recorded in an audit log.	**Both**
1.13	Expiry and batch tracking. The system SHALL support an optional expiry or best-before date for applicable products and, where needed, individual stock batches. WHEN stock is received, the owner or an authorised staff user SHALL be able to record quantity, receipt date, expiry / best-before date and batch details. The system SHALL NOT assume an expiry date or claim a product is near expiry unless the date is recorded or confirmed by an authorised user.	**MD**

ID	Requirement	Source
1.14	Near-expiry alerts. WHEN a product or batch approaches its recorded expiry date, the Inventory Tracker SHALL alert the owner using a configurable warning window. The alert SHOULD show product name, batch, remaining quantity, expiry date, days remaining and available sales / movement information. Near-expiry stock SHALL be distinguished from ordinary dead stock and low-stock alerts, and these alerts SHALL complement, not replace, the slow-moving and dead-stock alerts.	**MD**

Module 2: Analytics Dashboard

User story: As a shop owner, I want graphs, heatmaps and reports, so that I can identify patterns, peak times and profit performance.

ID	Requirement	Source
2.1	Peak and quiet time analysis. The system SHALL analyse transaction timestamps (and customer-traffic patterns) to identify the busiest and quietest hours, days and periods, and present them so the owner can plan staffing, break schedules and operations.	**Both**
2.2	Data sufficiency. Peak-period insights SHALL be based only on recorded data and the system SHALL clearly indicate when there is insufficient data for a reliable pattern.	**MD**
2.3	Product performance. For any selected period the system SHALL show most-sold products (Top 10), least-sold products, products with changing demand, and dead stock (no sales in X days).	**Both**
2.4	Income analytics. The system SHALL summarise income daily, weekly, monthly and yearly, together with total customer counts.	**PDF**
2.5	Profit and expense reports. Profit SHALL be calculated as total revenue minus total expenses for the selected period. The system SHALL provide weekly, monthly and yearly profit graphs, monthly net balance (profit or loss) and an expense breakdown by category.	**Both**
2.6	Heatmaps. The system SHALL provide an hourly customer heatmap, a day-wise footfall heatmap and a weekly sales heatmap.	**Both**
2.7	Standard charts. The dashboard SHALL include a line chart (income trend), bar charts (top products, customers per hour) and a pie chart (category-wise sales).	**PDF**
2.8	Seasonal trends. The system SHALL identify seasonal trends, correlate them with product categories, and support custom event analysis (festivals, weather impact).	**Both**
2.9	Repeat customers. The system SHALL track repeat visits, purchase preferences and items frequently bought by repeat customers, and identify high-value customers.	**Both**
2.10	Slow-moving alerts. WHEN a product has not sold for X days, the system SHALL alert the owner (for example, 'Salt 1kg has not sold for 8 days') and suggest discount, combo offer or in-store highlighting.	**Both**
2.11	Report export. The system SHALL provide daily, weekly, monthly and yearly digital reports and a one-click PDF export suitable for printing as a manual record.	**MD**
2.12	Data feed to AI. Revenue, expense, profit and loss figures SHALL be made available to the Business Health Check feature.	**MD**

Module 3: AI Business Insights

User story: As a shop owner, I want AI-powered insights and pricing suggestions, so that I can improve stock, pricing and sales decisions.

ID	Requirement	Source
3.1	Rule-based insights (V1, works offline). The system SHALL generate IF-condition tips without an external AI: fast-selling product notifications, product-not-sold warnings, best timing for offers, peak-time offer suggestions and a customer trends summary.	**PDF**
3.2	Dynamic pricing. WHEN demand changes, the system SHALL suggest price increases or decreases (for example, 'Cheese demand up 20%: increase price by Rs. 3').	**Both**
3.3	Inventory forecasting. The system SHALL predict when a product will go out of stock, recommend reorder quantities, and show expected profit and highest-profit items.	**Both**
3.4	Festival and seasonal prediction. WHEN seasonal events approach, the system SHALL forecast demand for relevant categories (Diwali: sweets, diyas, snacks; monsoon: umbrellas, tea; summer: cold drinks, ice cream).	**Both**
3.5	Combo suggestions. The system SHALL suggest profitable bundles (for example, Oreo + Milk, Bread + Jam, Soap + Shampoo).	**Both**
3.6	Auto offer generator. Based on stock and demand, the system SHALL suggest offers such as Buy 2 Get 1, flat 10%, or happy hours.	**PDF**
3.7	Competitor pricing (optional). WITH the owner's location permission, the system MAY compare prices of similar nearby stores and suggest competitive pricing or combos.	**PDF**
3.8	Social media trends. The system SHOULD surface trending items from social media relevant to the shop type.	**PDF**
3.9	Business Health Check. WHEN requested, the system SHALL analyse revenue, expenses, profit/loss margin, product movement, stock levels, peak periods and customer engagement, explain where the business is lagging, and suggest offers, schemes or strategies.	**MD**
3.10	Offer and scheme logic. WHEN suggesting an offer or scheme, the system SHALL consider current business position, available budget, profit margin if the offer is applied, likely customer benefit, target customer group (for example, ladies) and the intended outcome.	**MD**
3.11	Missing data. WHEN data needed for a recommendation is missing or insufficient, the system SHALL state the limitation instead of presenting an unsupported conclusion.	**MD**
3.12	Suggestion only. Pricing changes, offers, schemes and strategies SHALL be presented as suggestions. The system SHALL NOT apply any of them without explicit authorisation from the owner or an authorised administrator.	**MD**
3.13	Shop-type-aware personalisation. The system SHALL capture the business type at registration (existing shop-type dropdown) and use it to configure AI recommendations, market-trend discovery, product suggestions, marketing content, analytics and Business Health Check context, as well as stock and reorder recommendations, pricing insights, offers and campaigns. The system SHALL adapt to the selected category and subcategories and SHALL NOT be limited to the examples in this document.	**MD**
3.14	Shop-type examples. Clothing / fashion: trending garment types, styles, colours and categories. Electronics / mobile: smartphones, earbuds, portable chargers, accessories and other applicable categories. Grocery / general store: packaged food, household and seasonal products, such as ice cream where supported by current evidence. (SHOULD)	**MD**
3.15	Reviewing shop type. The owner SHALL be able to review and, where supported, correct the shop type and product categories used for personalisation. WHEN the shop type changes, future recommendations SHALL update without silently changing confirmed business records or stock data. The system SHALL indicate when trend data is unavailable, stale or insufficient instead of presenting unsupported claims as current market facts.	**MD**

ID	Requirement	Source
3.16	Current market trends. WHEN trend information is available, the system SHALL identify products and categories trending in the owner's business category, using relevant accessible sources such as authorised retail / trend data providers, reputable market reports, public trend information and supported social-media trend sources. This extends 3.8.	**MD**
3.17	Trend presentation. A trending-product suggestion SHOULD show the product or category, the observed trend, source(s), retrieval / update date and any confidence or evidence indicator; explain why it may suit the shop (shop type, sales history, seasonality, local relevance, current stock); and let the owner open the source link where available. External market trends SHALL be distinguished from the shop's own sales data.	**MD**
3.18	Trend honesty and control. The system SHALL NOT claim a product is trending unless source data supports it, and SHALL disclose source failure or outdated data. Suggested new products and stocking decisions SHALL be recommendations only: no stock record, supplier order or inventory change without confirmation from the owner or an authorised user.	**MD**
3.19	Market price comparison. WHEN reliable comparison data is available, the owner SHALL be able to compare a selected product's selling price with relevant market prices from authenticated or reputable sources (authorised retailer / catalogue feeds, recognised price-comparison providers, manufacturer or brand listings, other category-appropriate sources). Each displayed price SHALL identify its source, retrieval date / time and the applicable product or variant where available.	**MD**
3.20	Like-for-like comparison and display. The system SHALL compare like-for-like products where possible (model / SKU, brand, size, capacity, pack quantity, condition, accessories, warranty and other material attributes). The system SHOULD show a chart or table placing the owner's price beside sourced prices with source labels and the range or reference price used, and MAY show absolute and percentage difference when the owner's price is lower and the values are comparable.	**MD**
3.21	Evidence integrity of prices. The system SHALL distinguish source-backed prices from estimates, owner-entered prices and AI suggestions, and SHALL NOT label a price an 'authenticated market rate' without supporting evidence. It SHALL disclose limitations (unavailable sources, variant differences, old prices, location differences, taxes, delivery charges, discounts, stock availability). It SHALL NOT fabricate competitor prices, retailer names, citations, source links, discounts, or claims such as 'lowest price' or 'current market rate'.	**MD**
3.22	Owner review of comparisons. The owner SHALL be able to review, edit or reject a comparison and any proposed marketing claim before publication or sharing. Comparisons and price-based marketing remain subject to the existing owner-approval and publishing controls.	**MD**
3.23	Near-expiry offers. WHEN stock is unsold or slow-moving and has a recorded near-expiry date, the system SHOULD suggest a permitted discount, bundle, in-store highlighting or limited-time campaign. Examples such as 'Buy 1 Get 1' SHALL be proposed only where suitable for the product, remaining shelf life, stock quantity, budget and margin. The system SHALL show the expected financial impact where cost, price and quantity data are sufficient and state when data is insufficient. It SHALL NOT automatically discount, bundle, dispose of or reprice stock.	**MD**

Module 4: AI Chatbot and Voice Assistant

User story: As a shop owner, I want to ask questions and make supported updates by text or voice, so that I can get information quickly without navigating every screen.

ID	Requirement	Source
4.1	Stock queries. WHEN the owner asks about a product's stock (for example, 'how many Oreo biscuits are left?'), the chatbot SHALL retrieve and present the latest quantity.	**MD**
4.2	Business metric queries. WHEN the owner asks for sales, expenses, profit/loss margin or best sellers, the chatbot SHALL retrieve the data and respond concisely.	**MD**
4.3	Strategy queries. WHEN the owner asks how to increase sales, the chatbot SHALL run the Business Health Check analysis in the background and return a quick overview of findings and suggestions.	**MD**
4.4	Natural-language updates. WHEN the owner states an update (for example, '40 Oreo packets received today'), the agentic AI SHALL extract product and quantity and present the proposed change. The change SHALL be saved only after the user reviews and confirms it, unless a separately authorised workflow explicitly permits the action.	**MD**
4.5	Voice Log. WHEN a user provides a voice log, the system SHALL convert it to a structured proposed record (product update, stock receipt, expense or other supported entry), show the interpreted details for correction, and on saving associate it with the authenticated user and a timestamp.	**MD**
4.6	Permissions. The chatbot SHALL use the same permissions and shop scope as the signed-in user and SHALL NOT expose data the user is not authorised to see.	**MD**
4.7	Limitations. WHEN data cannot be retrieved or is insufficient, the chatbot SHALL say so rather than invent a value.	**MD**
4.8	Auditability. Confirmed chatbot-driven changes SHALL be retained with user, action, affected record and timestamp.	**MD**
4.9	Future scope: hotels and cafes. The system MAY let customers or staff select items from an uploaded menu and place orders by voice or manual selection; submitted orders SHALL be recorded and available to authorised users for handling and reporting. The QR-menu workflow is specified in 6.6 to 6.8.	**MD**

Module 5: AI Marketing Creator

User story: As a shop owner, I want to create professional promotional material without design skills, so that I can promote offers on WhatsApp, Instagram and other platforms.

ID	Requirement	Source
5.1	Template posters (no-AI base). The system SHALL provide templates: Today's Offer, Weekend Sale, Limited Stock, New Arrival and festival themes. The owner selects a template, product and discount; the app renders a poster with product name, price, discount text and shop name/logo.	**Both**
5.2	Captions and copy. The system SHALL generate WhatsApp status text, Instagram story captions, hashtags and short marketing messages.	**Both**
5.3	Festival-wishing posters. The system SHALL create festival posters (Diwali, Independence Day, Christmas, sale events) that can include the business name or logo and the owner's configured Instagram ID and other social media identifiers.	**Both**
5.4	Performance-driven campaigns. WHEN content is based on business insights (for example, low engagement among ladies), the system SHALL use available analytics findings and clearly distinguish system suggestions from owner-provided campaign details.	**MD**
5.5	Voice-to-Ad. WHEN the owner speaks an idea (for example, 'promote fresh vegetables today'), the system SHALL generate a poster, caption, Instagram reel script, WhatsApp story and a video ad concept (optional 15-second video).	**Both**

ID	Requirement	Source
5.6	Shop Branding Kit. The system SHALL auto-generate a logo, brand colour theme, digital banner, 'Offer starts today' templates and poster templates matching the brand identity.	**PDF**
5.7	AI product photography. The system SHALL enhance product photos (background, lighting, sharpness, highlighting) for ads.	**Both**
5.8	AI ad image generator (advanced). The system MAY generate full promotional posters and festival-themed artwork using AI image generation.	**Both**
5.9	Owner review and publishing control. The owner SHALL be able to review, edit, regenerate, accept or reject every poster, caption, hashtag and text before sharing. The system SHALL NOT publish or send content to any platform without the owner's explicit action. Content SHALL use business details and an offer or campaign selected or approved by the owner.	**MD**
5.10	Price claims in marketing. WHEN a market comparison is used in a customer-facing poster, caption or campaign, the system SHALL use only the claim and evidence selected or approved by the owner and SHALL keep the necessary source and date context in the content or its supporting details.	**MD**
5.11	Near-expiry campaigns. WHEN a near-expiry offer is approved for marketing, the system SHALL be able to use the approved product, offer, price, validity period and shop details to create a matching poster and caption.	**MD**
5.12	Logo onboarding. WHEN the owner first opens the Branding Kit or marketing setup, the system SHALL ask whether the business already has a logo. If yes, the owner can upload or select it and preview it before saving. The approved logo SHALL be stored in the shop's branding profile and reused by default in posters, banners and festival creatives unless the owner chooses otherwise. The original upload SHALL be preserved and SHALL NOT be silently replaced by an AI-generated logo.	**MD**
5.13	Guided logo creation. WHEN the owner has no logo or wants a new one, the system SHALL offer a guided workflow that collects preferences by text and, where supported, voice, with predefined options plus a custom answer. It SHOULD ask about logo approach (simple, calligraphy, typography, icon), language or script, colours and theme, specific icons or symbols, and style references, and MAY ask follow-up questions in a way suitable for users without design experience.	**MD**
5.14	Branding brief and approval. Voice answers SHALL be transcribed and structured into a consolidated branding brief that the owner can correct and MUST confirm before the logo is generated. The system SHALL then show a logo proposal; the owner can accept, reject, request changes or regenerate. A generated logo SHALL become the default branding asset only after explicit approval by the owner or an authorised administrator.	**MD**
5.15	Branding persistence. The owner SHALL be able to update the logo or branding preferences later while the current approved version stays in use until a replacement is confirmed. In multi-shop accounts each shop's branding profile SHALL stay tied to its own shop. After approval the owner SHALL be able to proceed to poster creation without repeating the questionnaire.	**MD**
5.16	Context-aware campaign sources. The system SHALL support poster and campaign generation from owner-provided offers, discounts and festival schemes; approved slow-moving or near-expiry suggestions; approved trending-product or new-arrival suggestions; approved market-price comparisons; and relevant Business Health Check findings. A selected suggestion SHALL carry its product, available details and approved offer into the poster workflow, and a near-expiry product MAY be proposed as a campaign with quantity, approved discount or bundle, validity and saved branding.	**MD**
5.17	Campaign by text or voice. WHEN the owner describes a festival or campaign by text or voice (for example, an Independence Day offer on 15 August), the system SHALL extract the event, product or category, offer terms and validity, and present the interpreted details for confirmation. The system SHALL ask the owner to provide or confirm essential missing information (discount, products, dates, terms, price) rather than inventing it.	**MD**

ID	Requirement	Source
5.18	Campaign output and integrity. Campaigns SHALL use the saved logo, brand colours and identity by default, plus business name and configured social IDs, and MAY include captions, hashtags, WhatsApp status text and Instagram story content. The owner can preview, edit, regenerate, accept or reject assets, and nothing is published or sent without explicit owner action. The system SHALL NOT invent stock, expiry dates, prices, discount terms, market comparisons, trend evidence or business details, and SHALL distinguish source-backed information, owner-provided terms and AI creative suggestions where material.	**MD**

Module 6: Business Management Tools

User story: As a shop owner, I want invoicing, loyalty, payments and multi-shop tools, so that I can serve customers better and run all my shops from one app.

ID	Requirement	Source
6.1	Invoices. WHEN a sale completes, the system SHALL generate a GST-compliant PDF invoice with business branding, save it automatically to history (with customer name and order), and allow print, WhatsApp share and email share.	**Both**
6.2	Customer loyalty. The system SHALL provide a customer QR code, reward points per purchase, automatic discounts and coupons, birthday wishes with special offers, SMS/WhatsApp offers (sent only with the owner's authorisation), and a customer database with purchase history and loyalty status.	**Both**
6.3	Cashbook and UPI. The system SHALL integrate with UPI payments (Google Pay, PhonePe, Paytm, BharatPe), track cash-in, cash-out and UPI logs, keep a daily ledger summary and auto-generate a day-end closing report.	**Both**
6.4	Multi-shop. One owner SHALL be able to manage several shops, switch between them, and see consolidated dashboards comparing profit, customers and sales trends and identifying best and worst locations. Each shop SHALL keep separate inventory and staff; suppliers MAY be managed centrally.	**Both**
6.5	Location filtering. WHEN a shop is selected, all data and reports SHALL be filtered to that location.	**MD**
6.6	QR menu setup (hotels, restaurants, cafes). WHEN the shop type is hotel, restaurant or cafe and hospitality ordering is enabled, authorised users SHALL be able to create or upload a menu (item names, descriptions, prices, availability). The system SHALL provide a QR code that opens that shop's menu and ordering interface on a compatible device. The QR menu and its orders SHALL belong to the correct shop, including in multi-shop accounts, and the workflow SHALL NOT be imposed on unrelated retail shop types.	**MD**
6.7	Customer ordering. The customer SHALL be able to browse the menu, choose items and quantities, and see an order summary (items, quantities, prices or charges where supported) before submitting. The system SHALL validate availability and current price before accepting an order and tell the customer about unavailable items or price changes. A submitted order SHALL be recorded with a unique order ID, shop ID, items, quantities, timestamp and status.	**MD**
6.8	Order handling. Submitted orders SHALL be available to authorised staff for review, preparation, handling and reporting. The system SHOULD support status updates (received, accepted, preparing, ready, completed) where enabled, SHALL enforce permissions for menu management, order handling and business records, and MAY support voice-based ordering consistent with Voice Log.	**MD**

4. Cross-Cutting Requirements

These rules apply across all modules.

A. Owner control of AI output

ID	Requirement	Source
X1.1	AI-generated prices, product details, stock suggestions, offers, schemes, marketing content and strategies SHALL be treated as suggestions until explicitly approved.	**MD**
X1.2	Each suggestion SHALL offer accept, edit and reject controls, and SHALL be saved or applied only after confirmation.	**MD**
X1.3	A later AI suggestion SHALL NOT overwrite a user-confirmed value without explicit approval.	**MD**
X1.4	The UI SHALL distinguish existing business data, user-entered data, externally sourced information and AI-generated suggestions wherever this affects a decision.	**MD**
X1.5	Recommendations SHALL NOT independently change prices, apply discounts, publish content or alter confirmed product or business data.	**MD**

B. Offline-first data persistence

ID	Requirement	Source
X2.1	WHEN there is no internet or the server is unavailable, the app SHALL keep working (full POS functions: sales, inventory updates, and other supported records) and store changes locally in Room / SQLite. No data SHALL be lost.	**Both**
X2.2	Locally stored records SHALL survive application restarts.	**MD**
X2.3	WHEN connectivity returns, pending changes SHALL sync automatically to Firestore without creating duplicate transactions, with a defined method to detect and resolve conflicts (including multiple users editing offline).	**Both**
X2.4	The app SHALL show pending, successful and failed sync states, and SHALL NOT mark a record as synced until the remote write is confirmed.	**MD**

C. Authentication and access

ID	Requirement	Source
X3.1	Login options: email and password, Google sign-in, OTP verification by SMS and call (where provider and region permit), and optional biometric authentication on compatible devices.	**Both**
X3.2	WHEN biometrics are unavailable or fail, a configured fallback method SHALL be offered.	**MD**
X3.3	Role-based permissions SHALL be enforced after login regardless of the login method.	**MD**
X3.4	Credentials SHALL be protected; raw OTPs and biometric data SHALL NOT be stored as ordinary application records.	**MD**

D. Multilingual support

ID	Requirement	Source
X4.1	The app SHALL offer a language selector; navigation, forms, status messages and help text SHALL appear in the chosen language, and the preference SHALL persist between sessions.	**MD**
X4.2	Chatbot and AI explanations SHALL respond in the selected language where feasible.	**MD**
X4.3	Product names, transaction records and other user-entered data SHALL stay accurate when the display language changes.	**MD**

E. UI / UX

ID	Requirement	Source
X5.1	Modern gradient theme, clean dashboard cards, animated charts, smooth navigation and light/dark mode.	**PDF**

F. Audit and security

ID	Requirement	Source
X6.1	All user actions, restricted-action attempts, role changes and confirmed AI-driven changes SHALL be logged for security and accountability.	**Both**

G. Source transparency and auditability for market intelligence

ID	Requirement	Source
X7.1	Source references, retrieval timestamps and product / category context for market-trend and price-comparison results SHOULD be retained where the source permits.	**MD**
X7.2	External market information SHALL be distinguished from the shop's own sales, inventory, supplier and customer records.	**MD**
X7.3	The system SHALL tell the user when a market-data source cannot be accessed, when information is stale, or when comparable data is insufficient.	**MD**
X7.4	The system SHALL NOT fabricate a source, citation, trend, competitor price, product claim or confidence score.	**MD**
X7.5	The owner's acceptance, edit or rejection of material AI recommendations and campaign proposals SHALL be recorded in the audit model.	**MD**
X7.6	Existing staff and shop-location access rules SHALL apply to market intelligence, branding assets, inventory alerts and campaign data.	**MD**
X7.7	Owner-approval controls SHALL apply to suggested product additions, reorder quantities, pricing changes, discounts, bundles, branding changes and generated campaigns.	**MD**
X7.8	Detailed design SHOULD define the data fields or collections for market-source records, price observations, product batches and expiry dates, branding profiles, logo versions, campaign source links and hospitality orders, while keeping the existing Firestore model.	**MD**

5. Technology Stack

Area	Choice
Platform	Android app (Kotlin) first. Optional later: web dashboard for PC (React / Next.js)
Navigation	Navigation Component with BottomNavigationView (4 tabs) plus floating chatbot
Authentication	Firebase Authentication: email + password, Google sign-in, phone OTP (SMS / call); Android biometric for local unlock
Cloud database	Cloud Firestore
Local storage	Room / SQLite with a sync queue (offline-first)
Charts	MPAndroidChart
AI layer	AI API called from a backend (Node / Python): text, vision, speech, image generation. Rule-based logic on-device for V1
Poster rendering	Android Canvas, or server-side image generation

6. Application Flow

Step	Screen / Stage	What happens
1	**Start / Login or Register**	Email + password, Google sign-in, OTP (SMS / call), optional biometric. Register: name, shop name, shop type, phone, email, password. Language selection.
2	**Main Dashboard**	Sales today, customers today, top-selling product, sync status, quick actions (Add Sale, Add Product), floating AI Chatbot / Voice button.
3	**Tab 1: Sales and Entry**	Add product (scan / photo / AI suggestion), sales entry, calendar, inventory, expiry dates and batches, expenses, suppliers.
4	**Tab 2: Analytics**	Income, profit and expense graphs, heatmaps, top/least products, dead stock, repeat customers, PDF export.
5	**Tab 3: AI Insights**	Rule-based tips, pricing, forecasts, combos, trending products for the shop type, market price comparison, near-expiry alerts and offers, Business Health Check, suggestion review (accept / edit / reject).
6	**Tab 4: AI Marketing**	Branding Kit onboarding (upload or create logo), poster templates, captions, festival posters, campaign builder from approved insights, voice-to-ad, product photo enhancer, review before sharing.
7	**Settings / Extra Tools**	Invoices, loyalty, cashbook and UPI, multi-shop switcher, staff roles, language, offline and sync status, security, QR menu and orders (hotel / cafe shops only).

7. Data Model (Firestore)

Collections from the original schema are kept. Collections marked (new) are added to support features that previously had no storage.

Collection	Fields
users	userId, name, email, phone, shopId(s), role, language, createdAt
shops	shopId, ownerId, shopName, shopType, address, instagramId, socialIds, logoUrl
products	productId, shopId, name, category, price, stockQty, lowStockThreshold, supplierId, imageUrl, barcode
sales	saleId, shopId, timestamp, customerCount, totalAmount, customerId, createdBy
saleItems	saleItemId, saleId, productId, quantity, price
expenses	expenseId, shopId, category, amount, date, createdBy
suppliers	supplierId, name, phone, nextDelivery, notes
supplierPayments (new)	paymentId, supplierId, amount, dueDate, paidDate, status
customers (new)	customerId, shopId, name, phone, birthday, loyaltyPoints, qrCode
invoices (new)	invoiceId, saleId, shopId, invoiceNo, gstDetails, pdfUrl, createdAt
cashbook (new)	entryId, shopId, type (cash-in / cash-out / UPI), amount, source, timestamp
campaigns	campaignId, shopId, imageUrl, caption, createdAt, status (draft / approved / shared), sourceLinks
aiSuggestions (new)	suggestionId, shopId, type, content, dataSource, status (pending / accepted / edited / rejected), decidedBy, decidedAt
voiceLogs (new)	logId, shopId, userId, transcript, parsedRecord, status, timestamp
auditLogs (new)	logId, shopId, userId, action, affectedRecord, timestamp
menus / orders (hospitality)	menuItemId, orderId, shopId, items, quantities, status, timestamp (QR-menu ordering for hotel and cafe shops)
productBatches (new)	batchId, productId, shopId, quantity, receiptDate, expiryDate, batchNo
marketSources (new)	sourceId, name, type, url, lastRetrieved, status
priceObservations (new)	observationId, productId, sourceId, price, variant, retrievedAt
trendRecords (new)	trendId, shopType, category, productOrCategory, sourceId, observedAt, evidence
brandingProfiles (new)	brandId, shopId, logoUrl, colours, brief, currentLogoVersion
logoVersions (new)	versionId, brandId, assetUrl, origin (uploaded / generated), approvedBy, approvedAt
Local only: syncQueue	Room table of pending changes: changeId, collection, payload, status, retryCount

8. UI / UX Blueprint

Principles: gradient modern theme, clean dashboard cards, MPAndroidChart graphs, smooth animations, light and dark mode.

#	Screen	Contents
1	**Login / Register**	Text inputs, gradient button, shop type dropdown, Google, OTP and biometric options.
2	**Language selector**	List of supported languages; preference saved to the profile.
3	**Home dashboard**	Total sales today, customers today, top product, sync indicator, quick actions.

#	Screen	Contents
4	**Product management**	Product list with stock indicators, add/edit screen, barcode and photo capture, AI suggestion review.
5	**Sales entry**	Date-time picker, product search, quantity stepper, customer count, bill summary, save button.
6	**Calendar**	Month view; tapping a date lists that day's sales.
7	**Analytics dashboard**	Line chart (income), heatmap (hourly customers), bar charts (top products), pie chart (category sales).
8	**AI insights**	AI cards with predictions, pricing tips, forecasts, combo suggestions; accept / edit / reject controls.
9	**Business Health Check**	Findings, margin and budget view, suggested offers and schemes with explanations.
10	**AI Chatbot / Voice**	Chat window with microphone; confirmation card before any saved change.
11	**AI marketing tools**	Template gallery, caption generator, voice-to-ad, branding kit setup, preview and approve screen.
12	**Extra tools**	Invoices, cashbook and UPI, loyalty, multi-shop switcher, staff management.
13	**Sync status**	Pending, synced and failed items; conflict resolution view.
14	**Trending products**	Trend cards with product or category, observed trend, source, date, evidence indicator and relevance to the shop.
15	**Price comparison**	Owner's price beside sourced prices (chart or table) with source labels, dates, variants and limitations.
16	**Expiry alerts**	Near-expiry list by product and batch, days remaining, suggested offers and expected impact.
17	**Branding Kit onboarding**	Has-logo question, logo upload and preview, guided questions (text or voice), brief confirmation, logo proposals.
18	**Campaign builder**	Pick an approved insight or describe a campaign by text or voice, confirm details, preview poster and captions.
19	**Menu and orders (hotel / cafe)**	Menu editor, QR code, incoming orders and status updates.
20	**Customer QR menu**	Customer-facing menu, cart, order summary and order confirmation.

9. Development Roadmap

Phase	Goal	Time	Scope
Phase 0	Scope and tech stack	1-2 days	Fix scope, platform (Android / Kotlin), Firebase, 4 main tabs plus chatbot, user roles. Write scope document.
Phase 1	Core setup, auth, offline base	4-6 days	Project with Navigation Component and bottom navigation. Firebase Auth: email, Google, OTP; biometric option. Registration with shop-type dropdown. Language framework. Room / SQLite and sync queue. Base Firestore design.
Phase 2	Sales and product entry	7-10 days	Product management, barcode scan, sales entry, stock deduction, customer count, calendar, expenses, suppliers, expiry dates and batches, staff roles and audit log.
Phase 3	Graphs and analytics	7-14 days	MPAndroidChart: income, peak time, top/least products, dead stock, profit, heatmaps, PDF export.
Phase 4	AI insights (rule-based first)	10-20 days	IF-condition tips, slow-moving alerts, near-expiry alerts and offers, shop-type-aware rules, rule-based Business Health Check, suggestion accept / edit / reject framework.
Phase 5	Marketing (templates first)	10-20 days	Canvas / server-rendered poster templates, festival posters with logo and social IDs, logo upload and branding profile, owner review screen.
Phase 6	Real AI and assistant	15-25 days	AI API via backend: captions, Business Health Check AI, product suggestions from photo, chatbot, Voice Log, voice-to-ad, guided logo creation, campaign builder, market trends and price comparison (with source records), multilingual responses.
Phase 7	Business tools and polish	Ongoing	Invoices, loyalty, cashbook / UPI, multi-shop, AI image generation, QR-menu ordering for hotels and cafes, dark mode, animations, performance and security hardening.

10. Release Scope

Release	Includes
V1 (launch)	Products, sales, inventory with expiry tracking and alerts, expenses, suppliers, roles; core analytics and heatmaps; rule-based insights; template posters with logo upload; shop-type selection at registration; offline-first with sync; email + Google login; basic invoice; multilingual framework.
V2	Real AI: Business Health Check, chatbot, Voice Log, AI product suggestions, captions, voice-to-ad; trending products and market price comparison with sources; guided logo creation and context-aware campaigns; OTP and biometric login; loyalty; cashbook / UPI; multi-shop; branding kit; competitor pricing.
Future	AI image generation, AI product photography upgrades, QR-menu and voice ordering for hotels and cafes, web dashboard for PC.

The original plan says V1 may launch without real AI, using rule-based tips. Real AI features (health check, chatbot, voice, image generation) arrive in V2 and later.

11. Open Questions

- Which languages will be supported at launch?
- UPI app sync (Google Pay, PhonePe, Paytm, BharatPe): confirm which integrations their APIs actually allow before committing scope.
- Which AI provider and which backend (Node or Python) will host AI calls, and what are the cost limits per shop?
- OTP by call depends on provider and region support; confirm availability for target regions.
- Should the chatbot be a floating button or a fifth tab?

- Which authorised trend and price-data providers (retail feeds, comparison services, social trend sources) will be used, and what are their licensing terms and costs?
- What default warning window should near-expiry alerts use, and should it vary by product category?
- Should the customer QR menu open as a web page (no app install) or inside the app? Which release should include it?

Appendix: What Changed in This Merged Version

Added from requirements_updated.md (missing in the original PDF)

- Business Health Check and offer / scheme logic
- AI Chatbot and agentic voice updates (Voice Log)
- AI-assisted product entry with owner confirmation
- Owner approval and human control over all AI output
- Google sign-in, OTP (SMS / call), biometric authentication
- Multilingual support
- Offline-first detail: Room / SQLite, duplicate prevention, conflict handling, sync states
- Audit logs; immediate role-change propagation; chatbot permission scoping
- PDF export of reports
- Peak-time staffing guidance and insufficient-data warnings
- Performance-driven marketing and social IDs on festival posters
- Hotel / cafe ordering (future scope)
- Supplier payment tracking; multi-shop separate inventory and staff

Added from StoreVisionPro2.pdf (missing in the original MD)

- Calendar view; editable sale date/time; edit / delete products; product photos
- Stock-out and pre-peak restock alerts; dead stock list; 'visited, no purchase' counter
- Income analytics; Top 10 lists; category pie chart; weekly sales heatmap
- Custom event analysis; high-value customer identification
- Rule-based offline AI tips for V1; auto offer generator
- Competitor price comparison; social media trend items
- Shop Branding Kit; richer voice-to-ad outputs (reel script, WhatsApp story, video concept)
- Loyalty extras (customer QR, birthday offers, SMS / WhatsApp offers)
- Cashbook and UPI specifics; daily closing report; invoice print and history
- UI/UX principles; tech stack; app flow; database schema; screen blueprint; roadmap

Inconsistencies fixed

- Orphaned 'Requirement 7' heading removed; staff-roles content merged into Module 1.
- Duplicate product-suggestion requirements (MD 2.6 and 2.7-2.10) merged into one set.
- AI pricing, forecasting and slow-stock advice moved out of the 'AI Marketing Generator' component into AI Business Insights.
- Database schema extended with collections for customers, invoices, cashbook, supplier payments, AI suggestions, voice logs and audit logs.
- Navigation clarified: 4 main tabs plus a floating chatbot and a Settings / Extra Tools area, replacing the earlier '5 modules vs 4 tabs' mismatch.
- Roadmap extended with auth, offline, chatbot, voice and multilingual work that the original roadmap did not schedule.

Added in this revision (from Requirements 17 to 24 of the updated MD)

- Requirement 17: shop-type-aware registration and AI personalisation (clothing, electronics, grocery examples, reviewable shop type)
- Requirement 18: current market trends with source, date and evidence; recommendations only
- Requirement 19: evidence-based market price comparison, like-for-like rules, price claims in posters, no fabricated prices

- Requirement 20: expiry and batch tracking, near-expiry alerts, waste-reducing offers with expected financial impact
- Requirement 21: guided Branding Kit onboarding and persistent shop logo (upload or guided creation, brief confirmation, approval)
- Requirement 22: context-aware poster and campaign generation from approved insights, by text or voice
- Requirement 23: QR-menu ordering for hotels, restaurants and cafes
- Requirement 24: source transparency and auditability for market intelligence
- Small wording gaps closed: chatbot saves only after confirmation unless a separately authorised workflow allows it; SMS / WhatsApp offers only with owner authorisation
- Supporting sections extended: glossary, data model (batches, market sources, price observations, trends, branding, logo versions), app flow, screens, roadmap, release scope and open questions